

Course contents:

1. Vector calculus : Review
2. Electrostatics
 - Coulomb's law and Electric fields
 - Divergence and Gauss's law
 - Electrostatic potential, Poisson's & Laplace's Equation
 - Capacitance and stored energy
 - Electric fields in matter
 - Boundary conditions
3. Magnetostatics
 - Magnetic fields
 - Fields due to current loop
 - Ampere's law
 - Inductance, Mutual inductance
 - Faraday's law
 - Magnetic energy
 - Generalized ampere's law
4. Maxwell's Equations and Wave equation

Reference Text Books

1. *Introduction to Electrodynamics*, David J Griffiths
2. *Engineering Electromagnetics*, W H Hayt, J A Buck
3. The Feynman Lectures on Physics, Volume II

Grading policy

- 20% Class tests (surprise quizzes included) + simulation assignment(s), if any
- 35% Mid-semester exam
- 45% End-semester exam